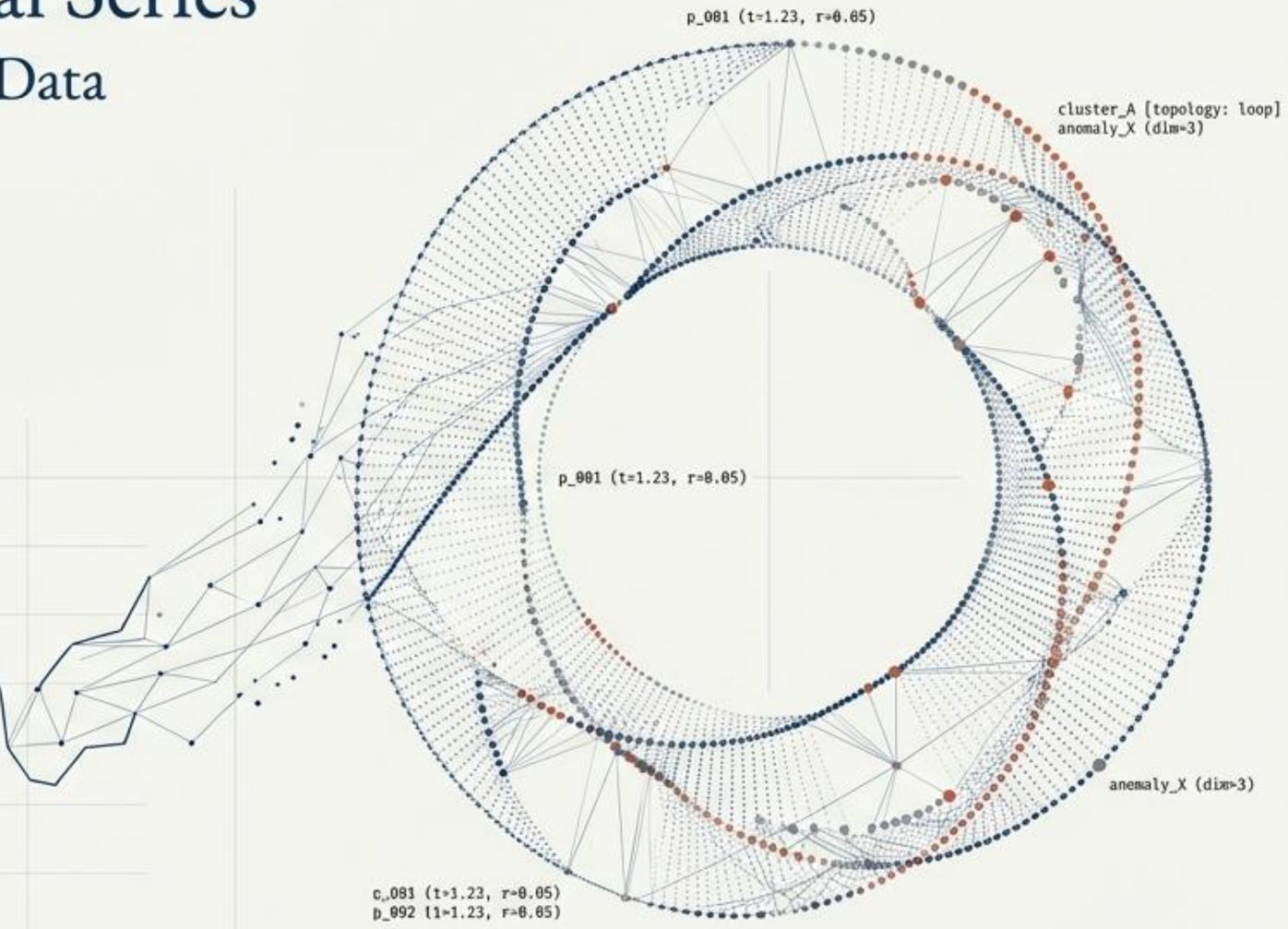
Introduction to Financial Series

Lecture 1: The Shape of Financial Data

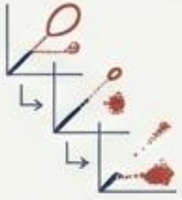
MPhil in Finance and Economics
University of Cambridge



UNIVERSITY OF CAMBRIDGE

NotebookLM

Financial stylized facts have direct topological counterparts

Classical Measure (Cont, 2001)	Topological Counterpart
Heavy Tails (Excess Kurtosis)	More persistent H_1 features (long arms form loops at larger radii)
Volatility Clustering (GARCH)	Time-varying persistence diagrams 
Leverage Effect (Asym. GARCH)	Asymmetric point cloud geometry
Zero Linear Autocorrelation	Symmetric point cloud

[R301A / F400 Curriculum Connection]

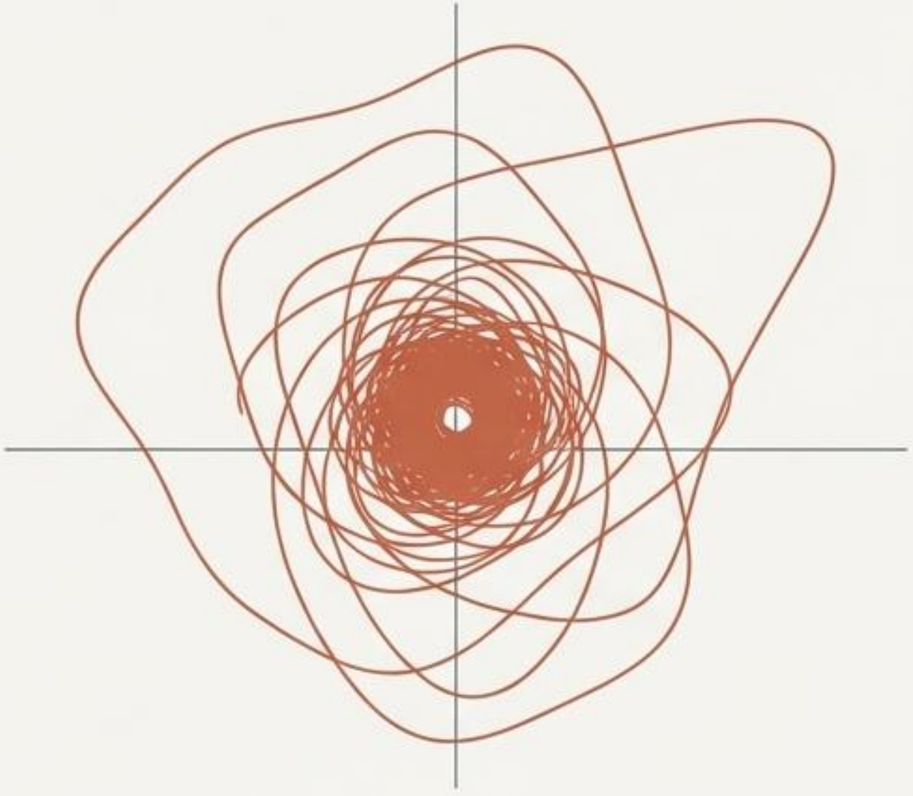
The persistence diagram serves as a non-parametric tail diagnostic. In a Gaussian point cloud, few persistent loops exist. Heavy tails produce outlier returns that create distinct geometric loops (H_1).

NotebookLM

Mean-reverting variance creates topological recurrence

$$dS_t = \mu S_t \, dt + \sqrt{v_t} S_t dW_{1,t}$$
$$dv_t = \kappa(\theta - v_t) \, dt + \xi \sqrt{v_t} dW_{2,t}$$

Mean-reversion speed (κ) and long-run variance (θ) force the variance path to revisit a central attractor.



Financial Interpretation

This recurrence structure appears as measurable H_1 features (loops) in the persistence diagram. TDA operates non-parametrically on the returns regardless of the exact generative volatility mechanism.

NotebookLM

Fractional differencing achieves stationarity while preserving geometric memory

$$(1-B)^d x_t = \sum_{k=0}^{\infty} \binom{d}{k} (-1)^k x_{t-k}$$

Search *d* over grid {0.05, 0.10, ..., 0.95} → Apply fractional differencing to the log-price series → Run ADF test on the result → Select the smallest *d* rejecting the unit root null at 5%

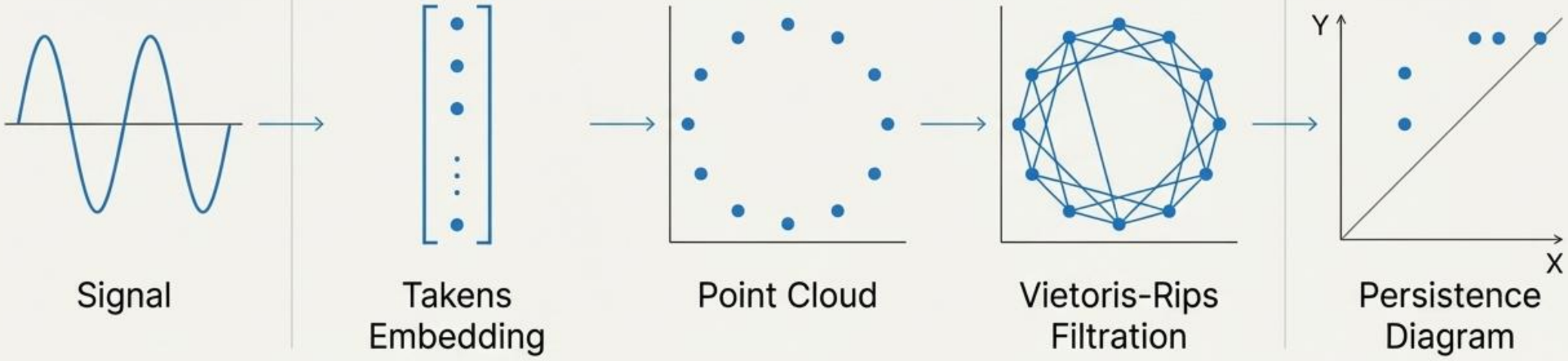
[E300 Curriculum Connection]

Standard differencing (d=1) destroys the long memory that carries topological signal. For FX series, optimal values hover around $d \approx 0.35 - 0.42$, retaining structure while satisfying stationarity.

NotebookLM

Delay coordinates reconstruct the topology of the original attractor

$$S_{W_{M,\tau}} f(t) = (f(t), f(t+\tau), \ldots, f(t+M\tau))^T$$



- By Takens (1981) theorem, the delay map preserves the topology of the original dynamic system.
- Example: Embedding a cosine wave with delay $\tau = P/4$ perfectly traces a circle in $\mathbb{R}^2$, producing a single dominant H_1 persistence feature.

NotebookLM

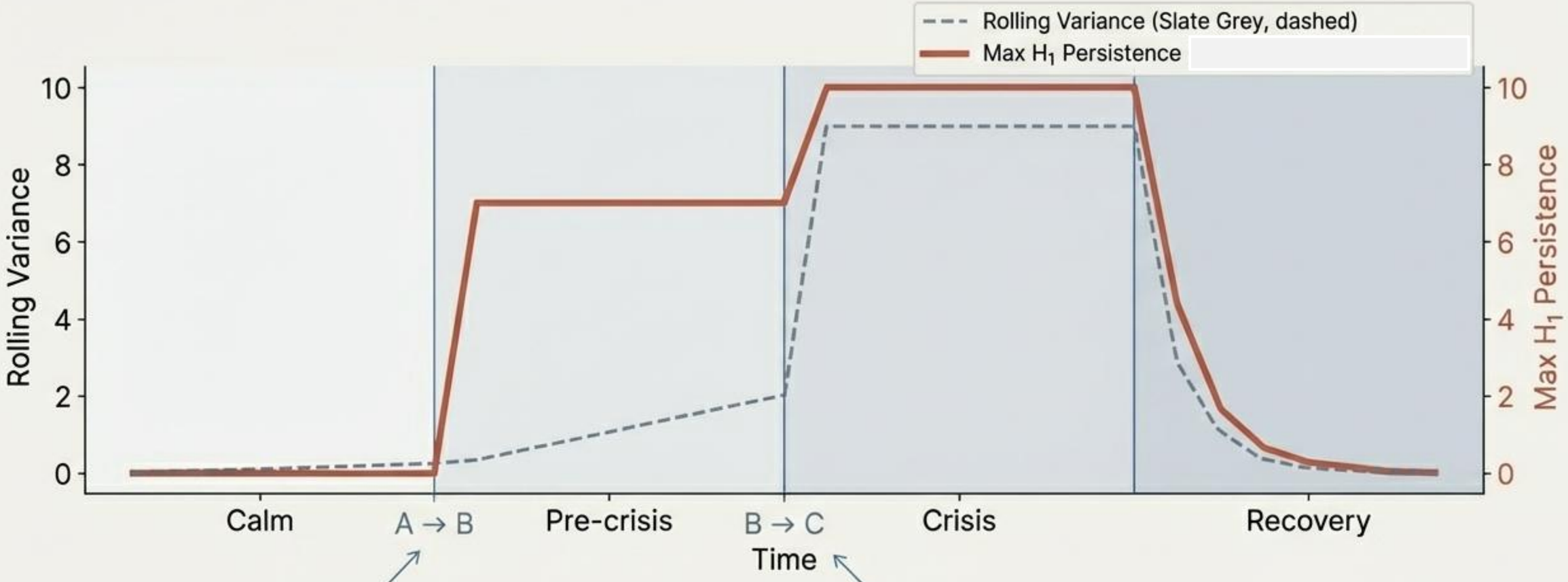
Constructing the sliding window embedding in Python

```python
def takens_embedding(data, delay, dimension):
    n = len(data)
    max_start = n - (dimension - 1) * delay
    if max_start <= 0:
        return np.zeros((1, dimension))
    indices = np.arange(dimension) * delay
    return np.array([data[i + indices] for i in range(max_start)])
```

This vectorized numpy implementation maps a 1D synthetic signal array into a multidimensional point cloud matrix, ready for the Vietoris-Rips filtration.

NotebookLM

TDA detects structural geometry changes that variance misses



Gradual Frequency Shift: Variance rises slowly. Max H_1 persistence shifts sharply and earlier, because changing periodicity directly alters the loop count.

Abrupt Amplitude Jump: Both metrics spike simultaneously. Variance functions purely as an amplitude scaler.

NotebookLM

Contagion is the geometric collapse of topological structure

Pairwise measures (like DCC-GARCH) report identical correlation matrices for Ring vs Full contagion.

Persistent homology easily distinguishes them: Ring contagion has a **persistent H_1 loop**, representing dangerous systemic feedback amplification that a **filled triangle kills**.

NotebookLM

Mapping LLM sentiment events into an 8-dimensional space

News Headline

Anthropic Claude API

$\mathbb{R}^8$ Vector

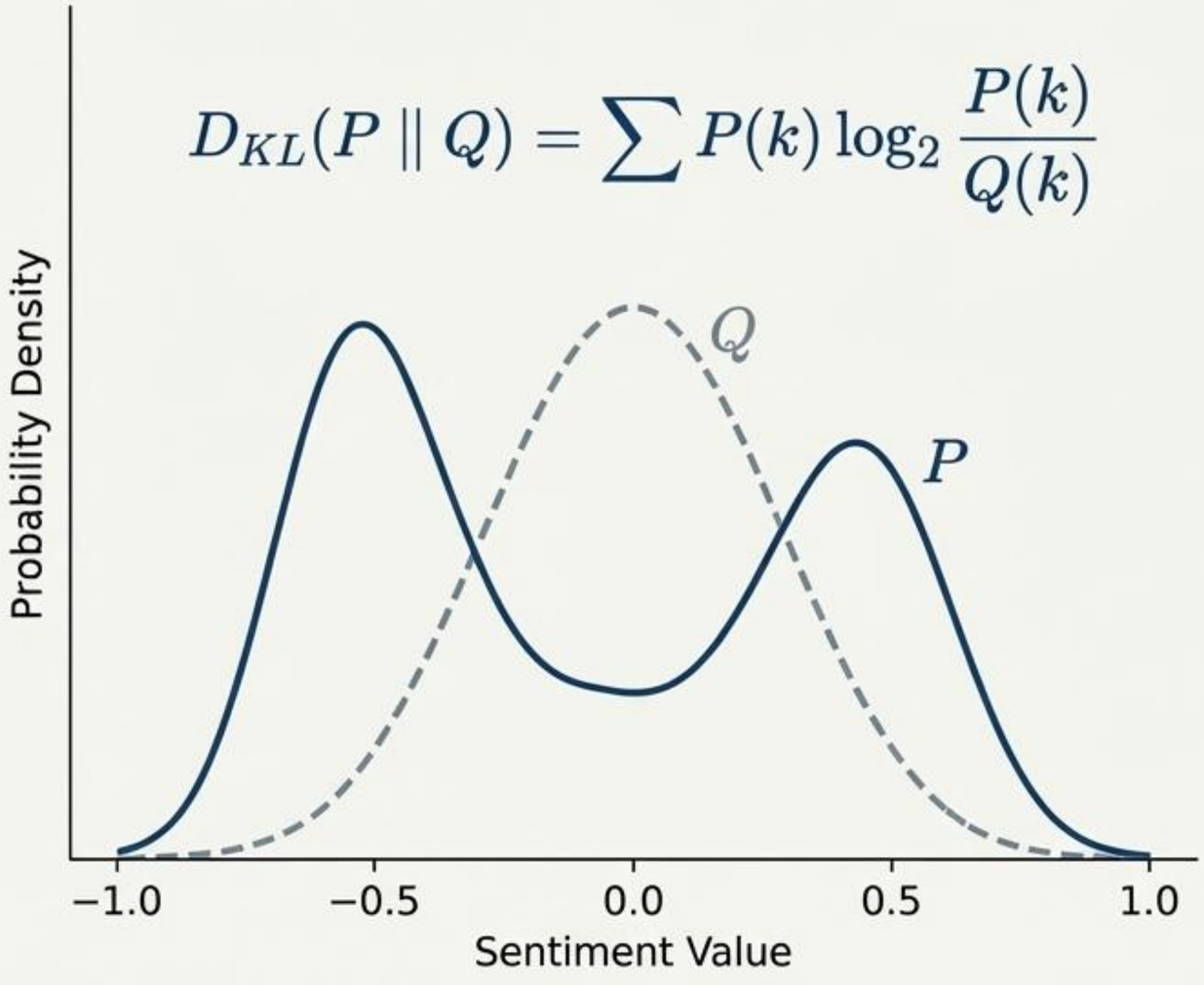
1. Normalised Time [0,1]
2. Net Sentiment (Pos − Neg)
3. Sentiment Magnitude
4. Country Encoding
5. Category Encoding
6. Significance (0.0 or 1.0)
7. Negative LM Score (calibrated)
8. Positive LM Score (calibrated)

[F550 Curriculum Connection]
The LLM labeller applies **prompt engineering** and **RAG** principles to score events strictly via **Loughran-McDonald** (2011) parameters, mixing API backfill with fixed anchor events (e.g., Lehman negative 0.98).

NotebookLM

KL divergence isolates shape shifts in the sentiment distribution

$$D_{KL}(P \parallel Q) = \sum P(k) \log_2 \frac{P(k)}{Q(k)}$$

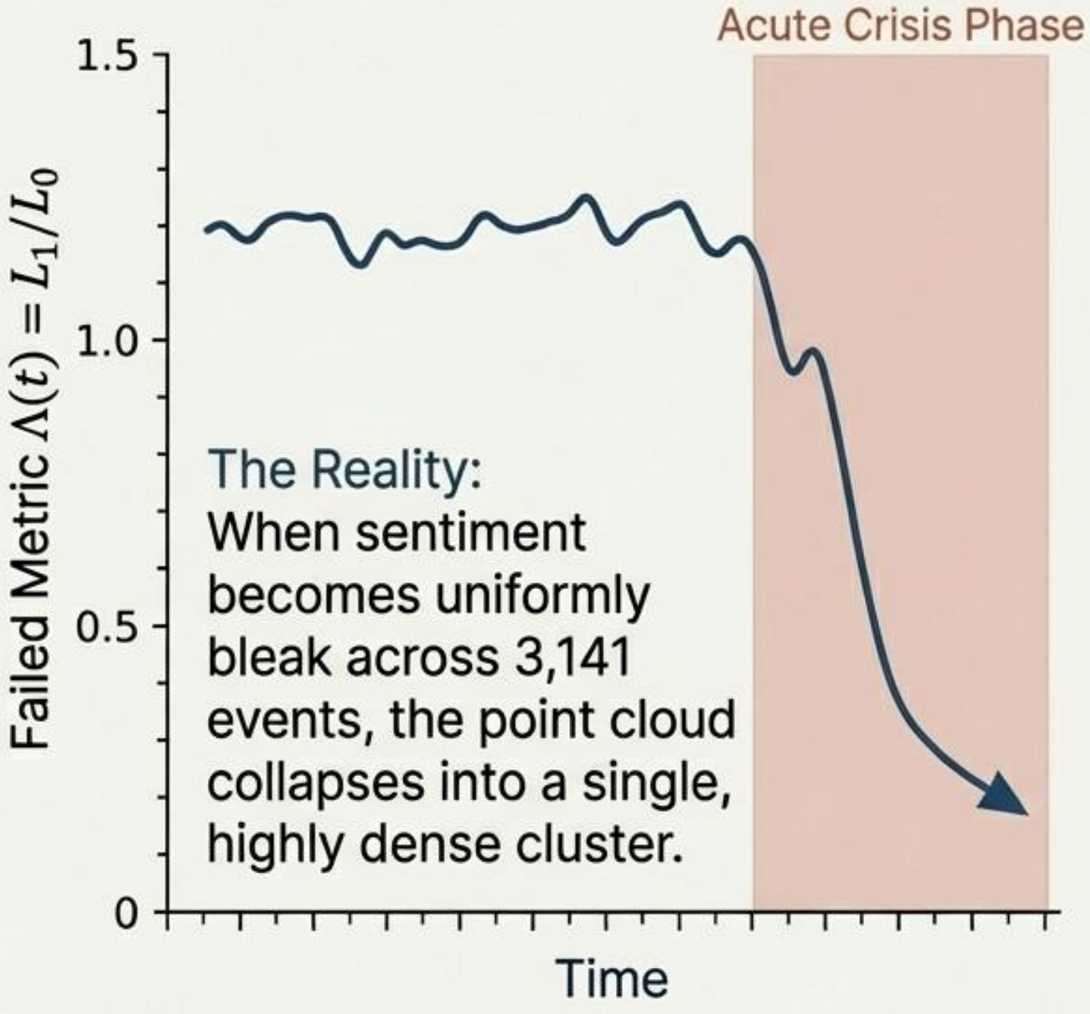


- Quantile binning normalises away pure level shifts. If mean sentiment shifts uniformly from -0.3 to -0.6, $D_{KL} = 0$.
- It fires only when the pattern fundamentally changes (e.g., shifting from broad market negativity to heavily concentrated bimodal shocks).
- Topology and KL Divergence are mathematically aligned: both are invariant to uniform scaling and detect geometric/distributional shape.

NotebookLM

The contagion ratio hypothesis: why L_1/L_0 failed on real data

The Hypothesis:
Contagion pathways form loops, so the ratio of loops (L_1) to clusters (L_0) should **rise** during systemic stress.



The Geometric Result:
Contraction kills loops ($L_1 \downarrow$) and strengthens the dominant component ($L_0 \uparrow$). Contagion is concentration, not dispersion.

NotebookLM

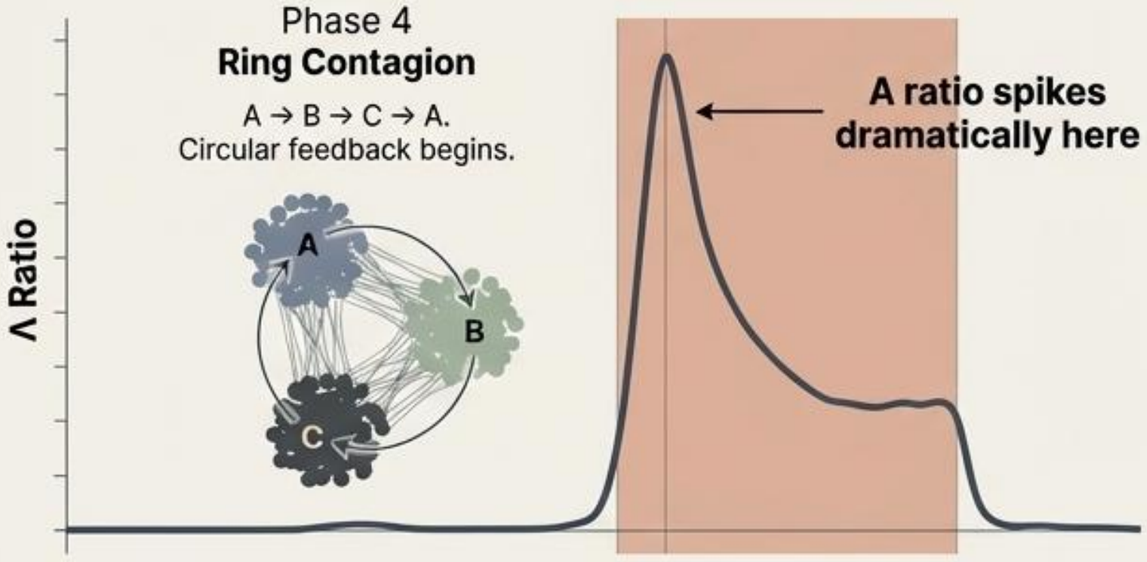
Contagion Theory Meets Empirical Reality

The Synthetic Theory

Key Metrics: 500 simulated events across 3 countries.

Mechanism: Predicted a 3-cluster transmission cycle (A → B → C → A).

Expected Signal: Loop ratio spikes as topological rings form.

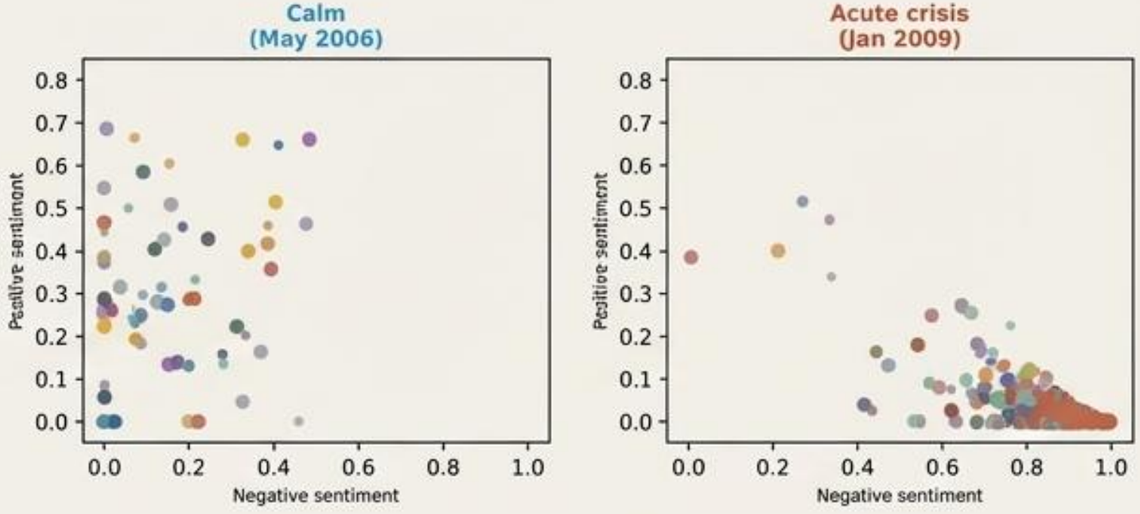


The Empirical Reality

Key Metrics: 3,141 real GFC sentiment events across 10 countries.

Mechanism: Sentiment uniformly collapses to high-negative, low-positive.

Actual Signal: Loops suppressed. A single dense cluster dominates.



The hypothesis failed productively: real crisis sentiment doesn't form complex transmission rings—it completely collapses into a single, dominant cluster.

NotebookLM

Inverting the Metric: The C(t) Crisis Detector

Track 1:
Raw Geometry
(Point Clouds)

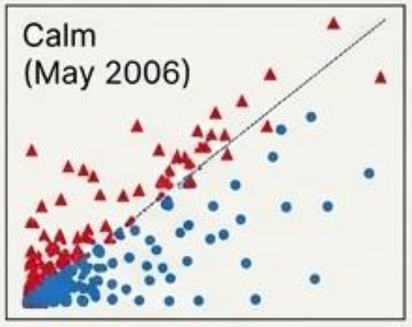
↓

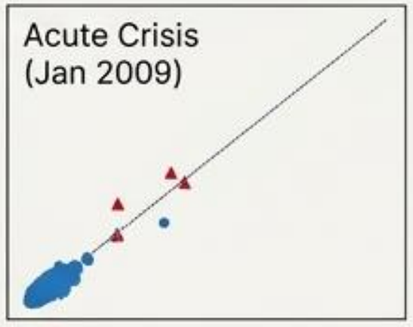
Track 2:
Topological
Signature
(Persistence)

↓

Track 3:
The Financial
Timeline
(C(t))

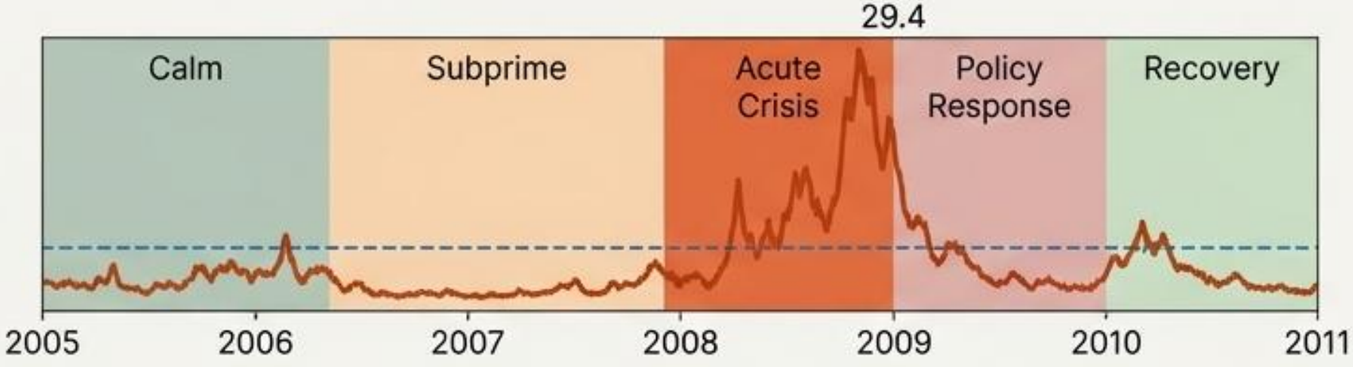
Dispersed Sentiment →
→ Concentrated Mass





H1 loops drop:
24 (Calm) → 14 (Acute Crisis)

L0 (cluster dominance) rises significantly as one cluster absorbs all persistence.



$$C(t) = L_0 / L_1$$

C(t) peaks at 29.4 in Nov 2008. EWA adaptive threshold successfully suppresses 2005-2006 calm-period noise.

By mapping this geometric contraction, the inverted metric provided a definitive topological crisis detector with a 7-month leading indicator (peaking Feb 2008) for the Global Financial Crisis.

NotebookLM

Topological concentration aligns perfectly with systemic crisis

$$C(t) = \frac{L_0(t)}{L_1(t) + \epsilon}$$

- Inverting the metric captures the true geometry of a market crash.
- C(t) rises 61% above baseline during the acute crisis.
- The total loop content (L_1) drops by 34%, confirming that circular topological pathways disappear under maximum systemic stress.

NotebookLM

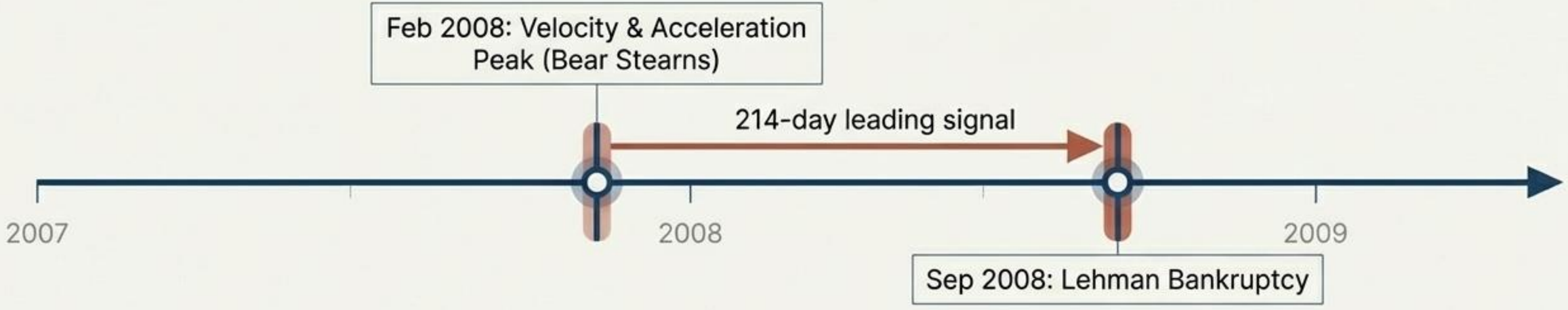
The derivatives of concentration provide severe early warnings

Velocity: dC/dt

$$\frac{dC}{dt} \approx \frac{1}{12h}\,[-25*C(t) + 48*C(t-h) - 36*C(t-2h) + 16*C(t-3h) - 3*C(t-4h)]$$

Acceleration: d^2C/dt^2

$$\frac{d^2C}{dt^2} \approx \frac{1}{12h^2}\,[35*C(t) - 104*C(t-h) + 114*C(t-2h) - 56*C(t-3h) + 11*C(t-4h)]$$

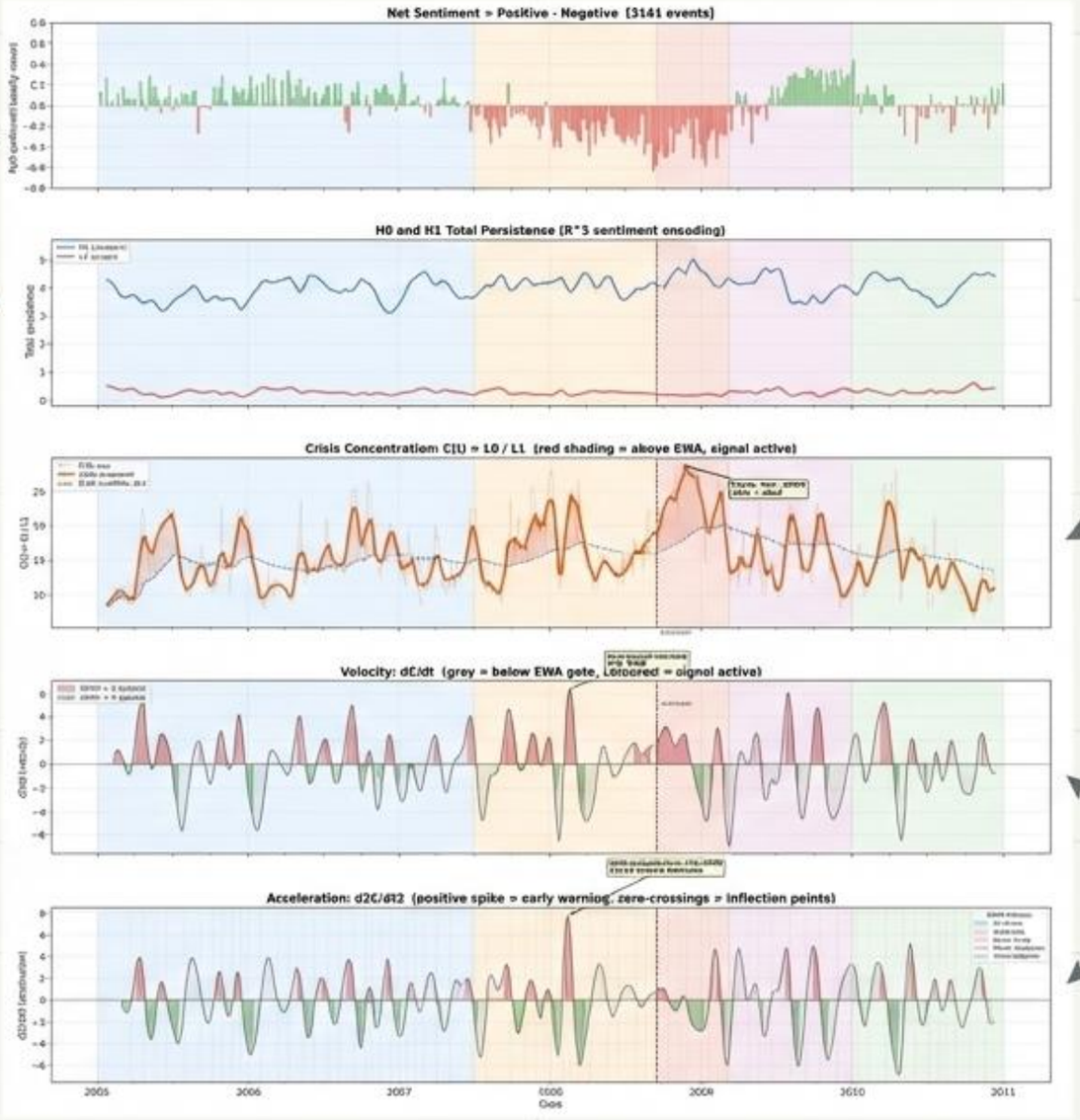


- C(t) is a concurrent indicator, but its derivatives flag accelerating concentration.
- EWA Gating: Requires an Exponentially Weighted Average gate to suppress calm-period noise, ensuring signals only fire when base concentration is elevated.

NotebookLM

Five-panel diagnostic: The geometry of the Global Financial Crisis

Raw weekly sentiment and the underlying H_0/H_1 persistence tracking the geometric collapse.



C(t) raw vs smoothed, showing the November 2008 peak.

The EWA-gated Velocity and Acceleration, highlighting the critical February 2008 early warning spike.

NotebookLM

Synthesizing connectivity, distance, shape, and balance

R_β: Connectivity Measures cluster mergers at a fixed scale. Requires calm baseline.	**R_W: Distance** Wasserstein shift from calm across all scales. Requires calm baseline.
R_{KL}: Shape 1D distribution shift. Self-contained.	**$C(t)$: Structural Balance** The L_0/L_1 dominance ratio. Self-contained.

The Forbes-Rigobon critique argues correlation artificially rises with volatility. $C(t)$ is entirely immune to this critique because it isolates pure geometric shape, independent of scale.

NotebookLM

Advanced Architectures: CNNs and Sequence Models

	Model Architecture	Input Type	Minimum Sample Size Constraint
Model Architecture	CNN	Persistence Image (L, 20, 20, 1). Learns spatial patterns in birth-persistence plane.	**Min. Obs: ~5,000** (Intraday/Long histories).
Input Type	LSTM	Topological summary sequence. Learns temporal regime memory.	**Min. Obs: ~2,000** (Daily with 10+ years).
Minimum Sample Size Constraint	Transformer	Topological summary sequence. Long-range attention for slow-build crises.	**Min. Obs: ~10,000** (High-frequency tick data).

Key Constraint Warning: Deep learning scales with data frequency. Fitting a Transformer to 750 monthly observations produces confident but meaningless predictions. XGBoost remains optimal for the 36-month GFC dataset.

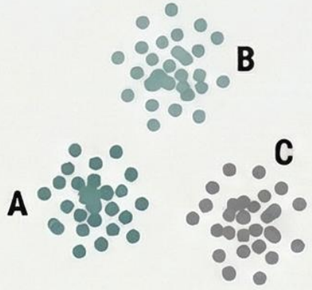
NotebookLM

The Topology of Markets

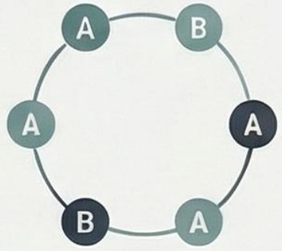
Comparing Theoretical Financial Contagion with Empirical Reality

THE SYNTHETIC MODEL (THEORETICAL FRAMEWORK)

Phase 1 to 3: From Calm to Hub Contagion
Markets begin as independent clusters (Days 1-200). Country A shifts to negative sentiment, eventually "Iefects" B, causing clusters to merge.



Phase 4: The Bing Contagion Hypothesis
Theery predicts chcular feedback path, creating a topological ring (HT1 feature).



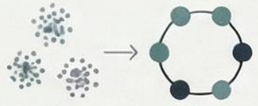
A Spike (Loops Form)
Causes the A ratio to spike, identifying emerging feedback loops.

TOPOLOGY CHANGE

Λ Ratio (L1/L0)
Tracks balance between loop (L1) and cluster (L0) persistence.

The Ring Contagion Hypothesis
Theory predicts circular feedback path, creating a topological ring (H1 feature).

VISUAL CHANGE



Dispersed → Ring

THE EMPIRICAL REALITY (GFC 2005–2010)

3,141 Sentiment Events Analyzed
Real-world data extracted and calibrated against the Loughran-McDonald financial sentiment dictionary.

Crisis Concentration, Not Loop Formation
During the 2008 Acute Criais, the point cloud did not form a ring but collapsed into a single, concentrated mass of high-negative sentiment.

The C(t) Ratio Inversion (L0/L1)
Because ampirical crises suppross loops, the formula was inverted to C(l)

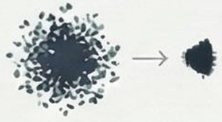
TOPOLOGY CHANGE

H1 COUNT DROPS / L0 RISES

3,141 Sentiment Events Analyzed
Real-world data extracted and calibrated against the Loughran-McDonald financial sentiment dictionary.

Crisis Concentration, Not Loop Formation
During the 2008 Acute Crisis, the point cloud did not form a ring but collapsed into a single, concentrated mass of high-negative sentiment.

VISUAL CHANGE

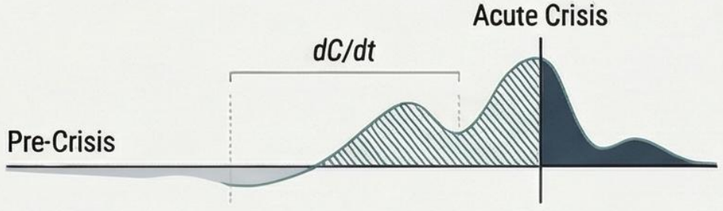


Dispersed → Concentrated

MODEL COMPARISON SUMMARY

PROPERTY	SYNTHETIC MODEL (THEORY)	EMPIRICAL MODEL (GFC DATA)
SCOPE	500 Events (3 Countries)	3,141 Events (10 Countries)
CRISIS SIGNAL	Λ Spike (Loops form)	C(1) Spike (Clusters dominate)
TOPOLOGY CHANGE	*H*1$ Features Appear	*H*1$ Count Drops / *L0*$ Rises
VISUAL CHANGE	3 Clusters → Ring	Dispersed → Concentrated
LEAD TIME	Concurrent with Phase 4	7-month lead via *dC/dt*

KEY FINDING: 7-MONTH PREDICTIVE LEAD TIME



By analyzing the rate of change (*dC/dt*) in market concentration, the model identifies crisis signals months before the acute phase.

NotebookLM